

Understanding file system structure (/ , /var, /etc,/home)? File &directory permission(chmod,chown,chgrp)? Sysmbolic &hard links

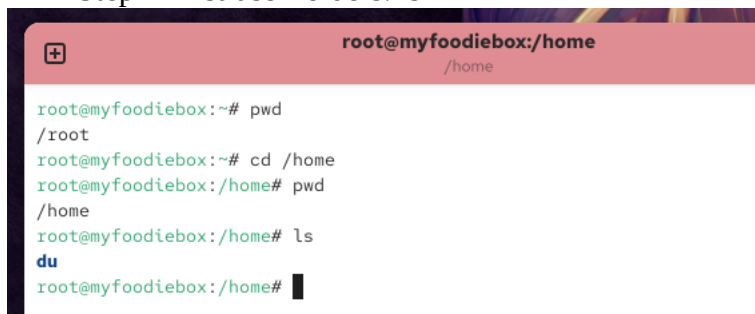
Q.1 Open a terminal and use the 'ls', 'cd', and 'pwd' commands to navigate to the /home directory, then list all user folders present. Take a screenshot of your terminal showing the commands and output.

Ans- Step 1- Check current directory: pwd

Step 2- Go to home directory: cd /home

Step 3- verify: pwd

Step 4- List user folders: ls



```
root@myfoodiebox:/home
root@myfoodiebox:~# pwd
/root
root@myfoodiebox:~# cd /home
root@myfoodiebox:/home# pwd
/home
root@myfoodiebox:/home# ls
du
root@myfoodiebox:/home#
```

Q.2 Create a new directory called 'myplaylist' inside your home folder, then use 'touch' to create three empty files named spotify.txt, youtube.txt, and wynk.txt inside it. Set the permissions so only you (the owner) can read and write these files, but others cannot access them.

Hint: Use 'chmod 600' on each file after creating them.

Ans- Step 1 – Go to your home directory: cd /home/student

Step 2- Create directory: mkdir myplaylist

Step 3- Enter directory: cd myplaylist

Step 4-Create files: touch spotify.txt youtube.txt wynk.txt

Step 5-Give owner read/write permission only: chmod 600 spotify.txt

chmod 600 youtube.txt chmod 600 wynk.txt

Step 6- Verify: ls

```
root@myfoodiebox:/home/du/myplaylist
/home/du/myplaylist

root@myfoodiebox:/# cd /home/du
root@myfoodiebox:/home/du# mkdir myplaylist
root@myfoodiebox:/home/du# cd myplaylist
root@myfoodiebox:/home/du/myplaylist# touch spotify.txt youtube.txt wynk.txt
root@myfoodiebox:/home/du/myplaylist# chmod 600 spotify.txt
root@myfoodiebox:/home/du/myplaylist# chmod 600 youtube.txt
root@myfoodiebox:/home/du/myplaylist# chmod 600 wynk.txt
root@myfoodiebox:/home/du/myplaylist# ls -l
total 0
-rw-----. 1 root root 0 Jun 29 10:10 spotify.txt
-rw-----. 1 root root 0 Jun 29 10:10 wynk.txt
-rw-----. 1 root root 0 Jun 29 10:10 youtube.txt
root@myfoodiebox:/home/du/myplaylist#
```

Q.3Change the ownership of the 'myplaylist' directory and all its files to another user on your system using the 'chown' command. If you are the only user, create a new user first.

Hint: Use 'sudo adduser newusername' to create a user if needed.

Ans- Step 1- Create Another User: useradd newuser

passwd newuser

Step 2-Change ownership: chown -R newuser:newuser /home/student/myplaylist

Step 3- Verify: ls -l/home/student

```
root@myfoodiebox:/
/

root@myfoodiebox:/# useradd newuser
root@myfoodiebox:/# passwd newuser
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
root@myfoodiebox:/# chown -R newuser:newuser /home/du/myplaylist
root@myfoodiebox:/# ls -l /home/du
total 0
drwxr-xr-x. 2 du du 6 Jun 29 08:39 Desktop
drwxr-xr-x. 2 du du 6 Jun 29 08:39 Documents
drwxr-xr-x. 2 root root 6 Jun 29 09:38 Down
drwxr-xr-x. 2 du du 6 Jun 29 08:39 Downloads
drwxr-xr-x. 2 du du 6 Jun 29 09:34 DU
drwxr-xr-x. 2 du du 6 Jun 29 08:39 Music
-rw-r--r--. 1 du du 0 Jun 29 09:10 my_linux.txt
drwxr-xr-x. 2 newuser newuser 60 Jun 29 10:10 myplaylist
drwxr-xr-x. 2 du du 6 Jun 29 08:39 Pictures
drwxr-xr-x. 2 du du 6 Jun 29 08:39 Public
drwxr-xr-x. 2 du du 6 Jun 29 08:39 Templates
drwxr-xr-x. 2 du du 6 Jun 29 08:39 Videos
root@myfoodiebox:/#
```

Q.4 Inside your home directory, create a symbolic link named 'favmusic' that points to the spotify.txt file you created earlier. Then, create a hard link named 'spotify_hardlink' to the same file. Show the output of 'ls -l' and 'ls -li' to demonstrate the difference between the links.

Ans- Step 1- Go to home directory: `cd /home/student`

Step 2- Create Symbolic link: `ln -s playlist/spotify.txt favmusic`

Step 3- Create hard link: `ln myplaylist/spotify.txt spotify_hardlink`

Step 4-Verify symbolic link : `ls -l`

Step 4-Verify hard link : `ls -li`

```

root@myfoodiebox:/home/du
root@myfoodiebox:/# cd /home/du
root@myfoodiebox:/home/du# ln -s /home/du/myplaylist/spotify.txt /home/du/favmusic
root@myfoodiebox:/home/du# ln myplaylist/spotify.txt spotify_hardlink
root@myfoodiebox:/home/du# ls -l
total 0
drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Desktop
drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Documents
drwxr-xr-x. 2 root    root     6 Jun 29 09:38 Down
drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Downloads
drwxr-xr-x. 2 du      du      6 Jun 29 09:34 DU
lrwxrwxrwx. 1 root    root     31 Jun 29 10:26 favmusic -> /home/du/myplaylist/spotify.txt
drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Music
-rw-r--r--. 1 du      du      0 Jun 29 09:10 my_linux.txt
drwxr-xr-x. 2 newuser newuser 60 Jun 29 10:10 myplaylist
drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Pictures
drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Public
-rw-----. 2 newuser newuser 0 Jun 29 10:10 spotify_hardlink
drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Templates
drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Videos
root@myfoodiebox:/home/du# ls -li
total 0
16777345 drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Desktop
16777346 drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Documents
33685662 drwxr-xr-x. 2 root    root     6 Jun 29 09:38 Down
33685633 drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Downloads
50331785 drwxr-xr-x. 2 du      du      6 Jun 29 09:34 DU
179 lrwxrwxrwx. 1 root    root     31 Jun 29 10:26 favmusic -> /home/du/myplaylist/spotify.txt
33685634 drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Music
155 -rw-r--r--. 1 du      du      0 Jun 29 09:10 my_linux.txt
174 drwxr-xr-x. 2 newuser newuser 60 Jun 29 10:10 myplaylist
50331778 drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Pictures
136 drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Public
175 -rw-----. 2 newuser newuser 0 Jun 29 10:10 spotify_hardlink
50331777 drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Templates
137 drwxr-xr-x. 2 du      du      6 Jun 29 08:39 Videos
root@myfoodiebox:/home/du#

```

Q.5 Suppose you want to share your 'myplaylist' folder with a group of friends on the same Linux system. First, create a new group called 'musiclovers' using the 'groupadd' command, add your user to this group, and then change the group ownership of 'myplaylist' to 'musiclovers' using 'chgrp'. Set the permissions so that only group members can read and write files in this directory, but others cannot.
Hint: Use 'chmod 770' on the directory after changing the group.

Ans- Step 1: Create group: `groupadd musiclovers`

Step 2- Add user to group: `usermod -aG musiclovers student`

Step 3- Change group ownership: `chgrp -R musiclovers /home/student/myplaylist`

Step 4- Set directory permission: `chmod 770/home/student/myplaylist`

Step 5- Verify: `ls -ld /home/student/myplaylist`



root@myfoodiebox:/

```
root@myfoodiebox:/# groupadd musiclovers
root@myfoodiebox:/# usermod -aG musiclovers du
root@myfoodiebox:/# chgrp -R mucislovers /home/du/myplaylist
chgrp: invalid group: 'mucislovers'
root@myfoodiebox:/# cat /etc/group | grep music
musiclovers:x:1002:du
root@myfoodiebox:/# chgrp -R muciclovers /home/du/myplaylist
chgrp: invalid group: 'muciclovers'
root@myfoodiebox:/# chgrp -R musiclovers /home/du/myplaylist
root@myfoodiebox:/# chmod 770 /home/du/playlist
chmod: cannot access '/home/du/playlist': No such file or directory
root@myfoodiebox:/# ls /home/du
Desktop    Down      DU         Music      myplaylist Public     Templates
Documents Downloads favmusic  my_linux.txt Pictures    spotify_hardlink Videos
root@myfoodiebox:/# chmod 770 /home/du/myplaylist
root@myfoodiebox:/# ls -ld /home/du/myplaylist
drwxrwx---. 2 newuser musiclovers 60 Jun 29 10:10 /home/du/myplaylist
root@myfoodiebox:/#
```